🚀 NASA Technical Manual QA

Ask questions about the NASA Systems Engineering Handbook

🔍 Enter your question

List different types of technical reviews and their purpose

Ask

💡 Answer

Based on the provided excerpts, here is a compiled list of different types of technical reviews and their purposes:

1. **System Requirements Review (SRR)**
   - **Purpose**: To ensure that system requirements are properly defined and aligned with the mission objectives. This review is aimed at confirming that the system requirements are complete, consistent, and feasible to meet.
2. **System Design Review (SDR)**
   - **Purpose**: To evaluate the proposed system design against the requirements and determine if the selected design is feasible and capable of meeting mission objectives.
3. **Preliminary Design Review (PDR)**
   - **Purpose**: To ensure that the preliminary design meets the defined technical requirements, is feasible, and supports the program's objectives. It evaluates the overall design approach, including architecture and design-to-cost.
4. **Program Management Council (PMC) Review**
   - **Purpose**: To approve and provide key management oversight during the program's lifecycle. It includes reviewing gate products for approval before transitioning to the next KDP (Key Decision Point).

5. **Phase-to-Phase Technical Reviews**
   - Typical Activities Conducted:
     1. Identifying, planning, and conducting phase-to-phase technical reviews.
     2. Establishing the purpose, objective, and entry and success criteria for each review.
     3. Forming the review team.
     4. Identifying and resolving action items resulting from the review.
6. **Key Decision Point (KDP) Reviews**
   - **Purpose**: To approve the transition between program phases (e.g., from Formulation to Implementation). CM (Configuration Management) plays a critical role in measuring readiness for a KDP, and outputs like Configuration Status Accounting (CSA) reports are reviewed to ensure compliance.
7. **Research and Technology Reviews**
   - **Purpose**: These reviews support management by evaluating progress in R&D efforts and ensuring research objectives align with broader program/project goals.
8. **Audit and Review of Configuration Management (CM) Results**
   - **Purpose**: To validate that configuration items, control measures, dispositions of change requests, and audit results meet the required standards and objectives. Outputs include corrective actions where necessary.

General Principles:

- Technical reviews aim to evaluate work products, confirm readiness, and support decision-making for the next phase.
- They capture key decisions made and the rationale behind those decisions.
- They are interconnected with risk management, decision analysis, and technical planning.

By systematically conducting these reviews, NASA ensures alignment with technical, programmatic, and mission goals throughout the project lifecycle.

📚 Sources

• 3.0 NASA Program/Project Life Cycle (Page 35)

- 4.4.1.2.1 Define Alternative Design Solutions (Page 78)
- 6.6.1.2 Process Activities (Page 178)
- 6.7.2.3 for more on technical reviews.) (Page 162)